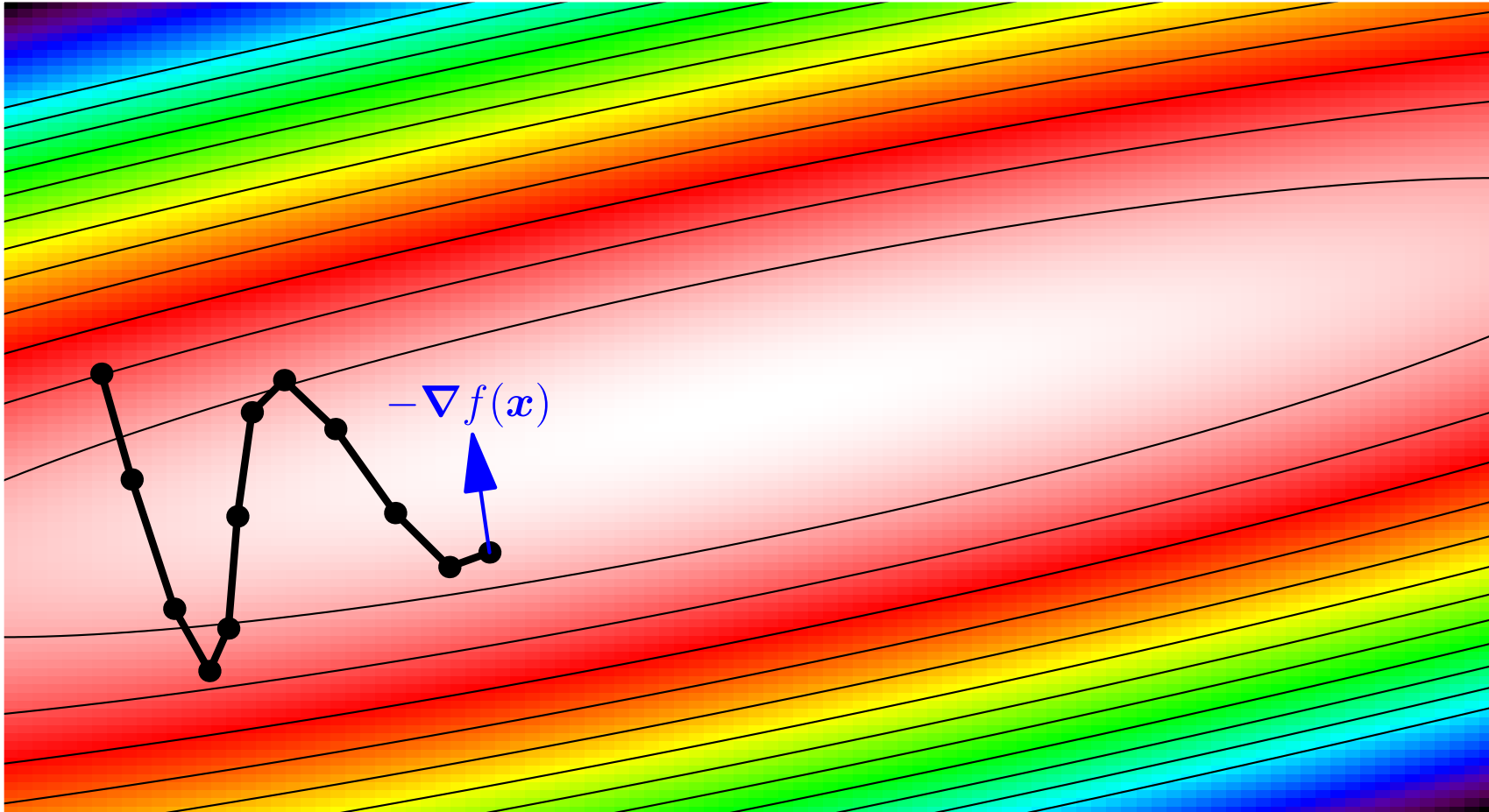


# Mathematics of Machine Learning

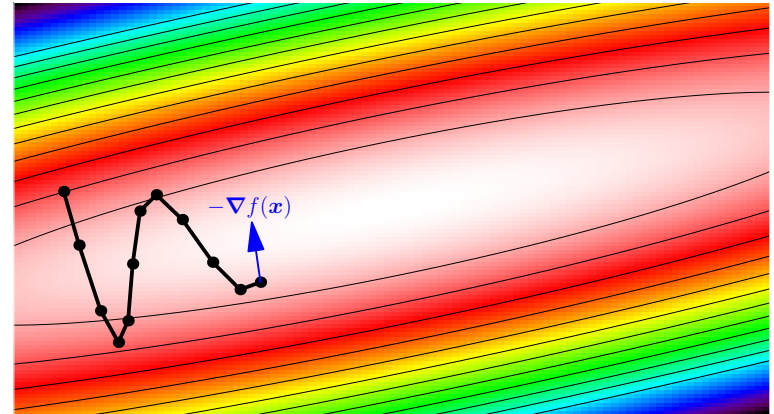
## *Stochastic Gradient Descent*



*SGD, momentum, step size, ADAM*

# Outline

1. **SGD**
2. Momentum
3. Loss Landscapes



# The Next Step

- In most machine learning problems we design a network and a loss function■
- The rest requires simple optimisation■
- Although Newton and Quasi-Newton methods ensure rapid convergence for many tasks this apparent advantage is delusionary■
- Since the raise of deep learning, gradient descent and its variants have become dominant■

# Mini-Batch Learning

- Rather than computing the gradient of the loss function, e.g.

$$\nabla L_{\mathcal{D}}(\mathbf{w}) = \nabla \sum_{(\mathbf{x}, y) \in \mathcal{D}} L(f(\mathbf{x}|\mathbf{w}), y)^2$$

- We often compute an approximation

$$\nabla L_{\mathcal{B}}(\mathbf{w}) = \nabla \sum_{(\mathbf{x}, y) \in \mathcal{B}} L(f(\mathbf{x}|\mathbf{w}), y)^2$$

- where  $\mathcal{B} \subset \mathcal{D}$  is a randomly sampled subset of the training set
- Usually  $|\mathcal{B}| \ll |\mathcal{D}|$

# Stochastic Gradient Descent

- Stochastic gradient descent (SGD) follows the gradient of mini-batches■
- Computationally this is efficient because it is much quicker to compute  $\nabla L_{\mathcal{B}}(\mathbf{w})$  than  $\nabla L_{\mathcal{D}}(\mathbf{w})$ ■
- The batch gradients add noise (are stochastic) as we are only sampling the dataset■
- This noise might help escape local-minima■(but I'm not sure there is compelling evidence for this)■
- Taking many smaller steps of approximate gradients reduces the likelihood of divergence■

# Automatic Differentiation

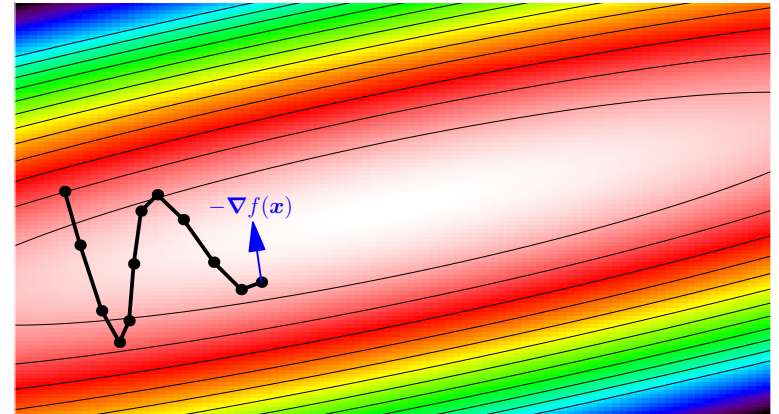
- For gradient descent we need to compute the gradient■
- For most of my life this was just a pain■
- You guys have it easy, modern deep learning frameworks do this for you automatically■
- This is a *game changer*!■ We are no longer afraid of large models■
- We can differentiate through algorithms allowing us to training incredible sophisticated networks■

# Convergence

- Asymptotic convergence of gradient descent (and SGD) is slower than quasi-Newton methods■
- But there are three reasons why we don't really care■
  - ★ For complex models we spend a lot of time before we are close to a quadratic minima where asymptotic convergence kicks in■
  - ★ These days we often use ReLUs (rectified linear units) which are non-analytic. The convergence we discussed depends on a Taylor expansion that assumes analyticity■
  - ★ We want to minimise the generalisation error, but are only minimising a poor surrogate, namely the training error■

# Outline

1. SGD
2. **Momentum**
3. Loss Landscapes

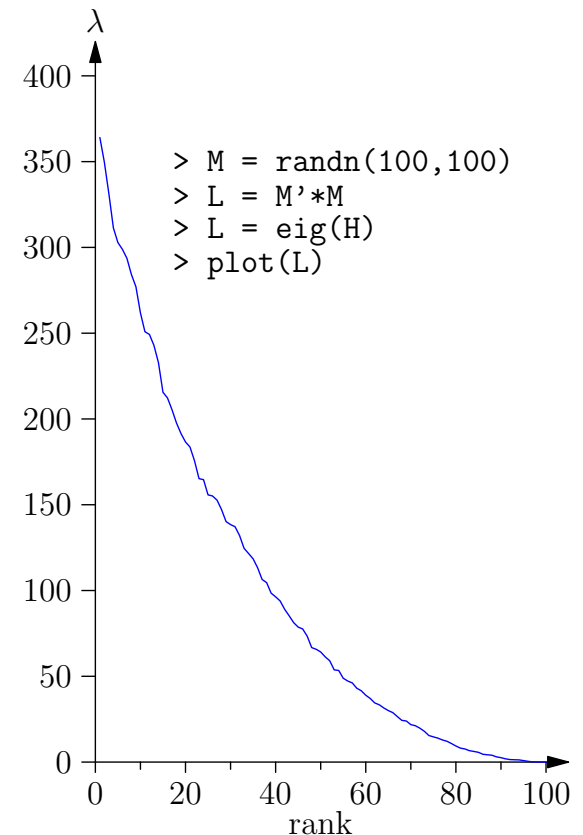


# Step Size

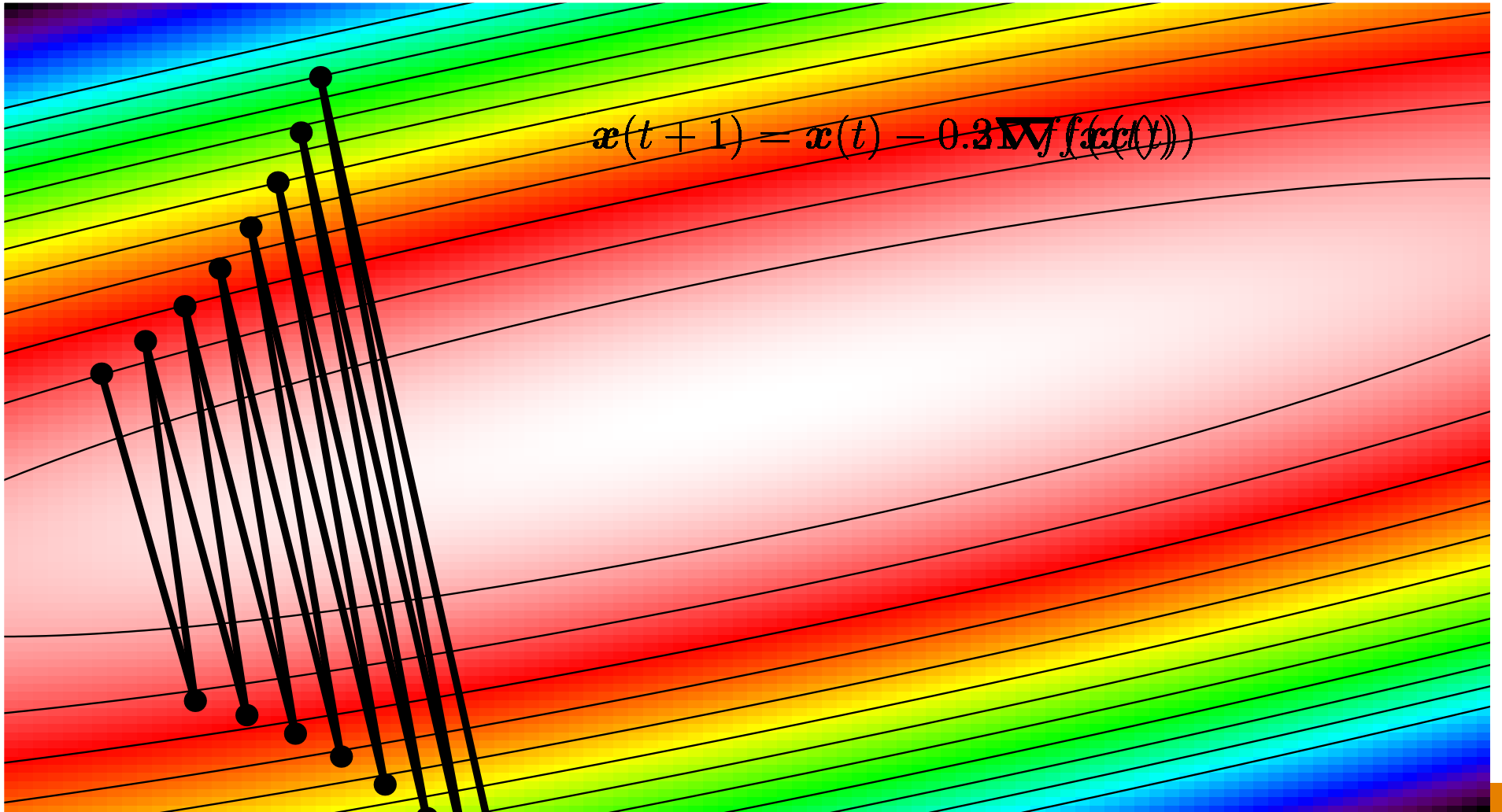
- The optimal step size depends on the gradient and the curvature (second derivative)■
- For a quadratic minima the minimum is given by

$$x^* = x - \frac{f'(x)}{f''(x)}, \quad \mathbf{x}^* = \mathbf{x} - \mathbf{H}^{-1} \nabla f(\mathbf{x}) \blacksquare$$

- In high dimensions the Hessian  $\mathbf{H}$  will have a spectrum of eigenvalues■
- This means there are different scales■



# Gradient Descent



# Avoiding Divergence

- We saw in the last lecture that gradient descent can actually diverge from a local optimum if you have a too big step size■
- If we use too large a step size we quickly get NaN errors■
- If we only use SGD the step size is determined by the largest eigenvalue of the Hessian■
- This seems impossibly slow■, but weirdly straightforward SGD is often used in deep learning■ (although nearly always with momentum)■

# Momentum

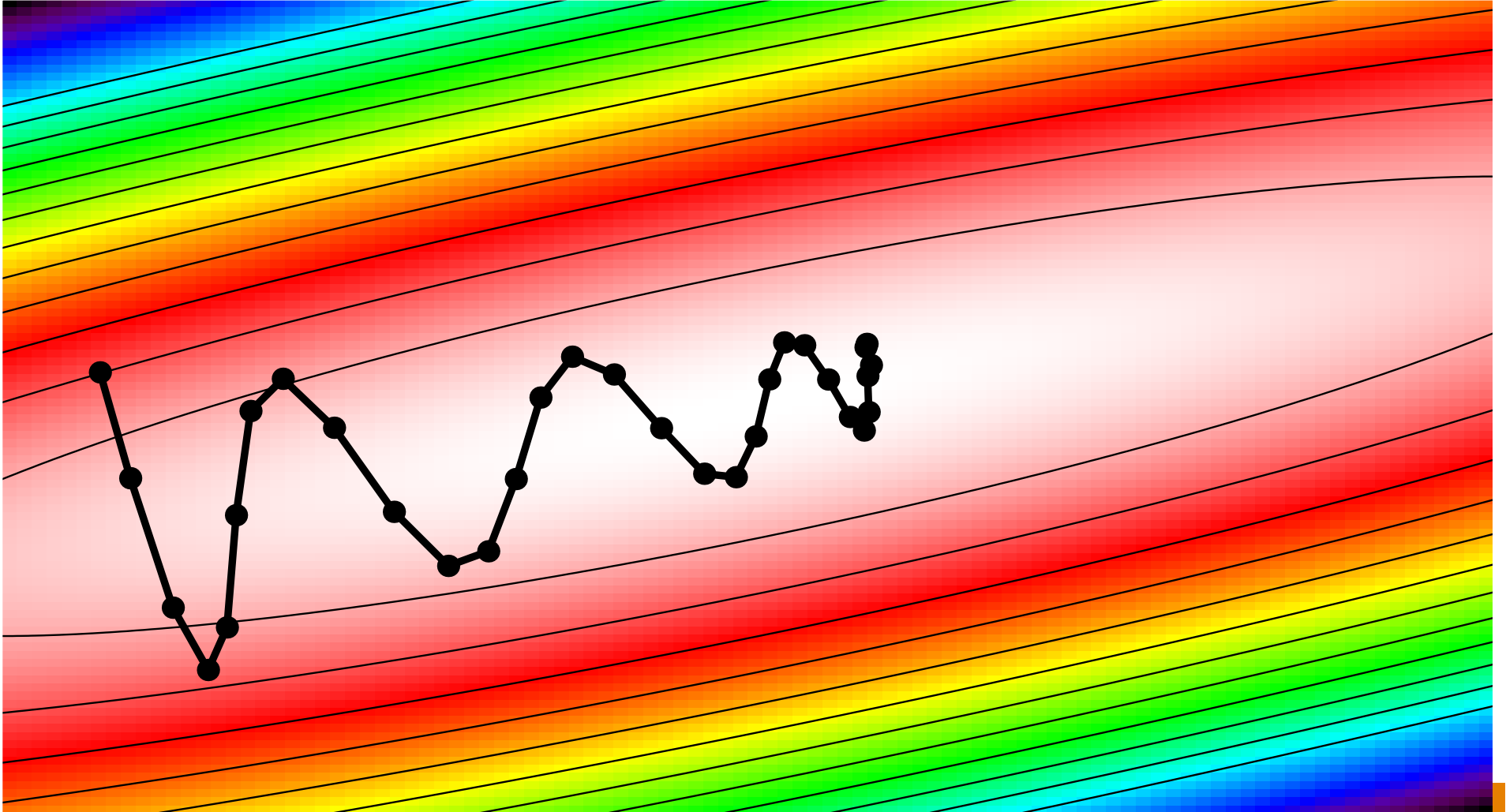
- In high dimensions we zig-zag:
  - ★ stepping consistently in directions with low curvature
  - ★ jumping backwards and forwards past the minimum in directions with high curvature■
- By introducing “momentum” we can increase our steps in low curvature directions and decrease it in high curvature directions

$$\mathbf{v}(t+1) = \gamma \mathbf{v}(t) - \eta \nabla f(\mathbf{x}(t))$$

$$\mathbf{x}(t+1) = \mathbf{x}(t) + \mathbf{v}(t+1)■$$

$0 < \gamma < 1$  is a damping term■

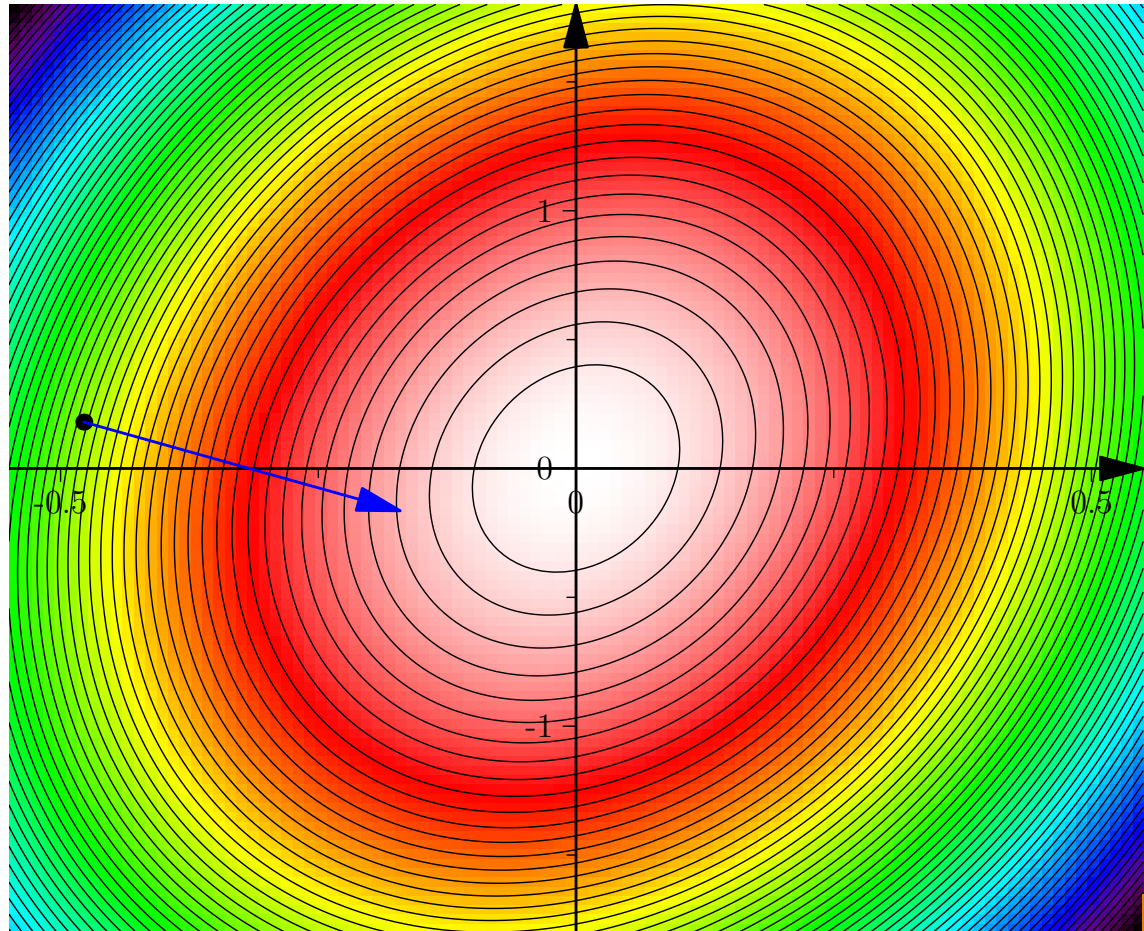
# Gradient Descent with Momentum



# Adaptive Methods

- A major difficulty of high dimensional optimisation is the existence of different scales■
- That is, there are some directions where we have to move a lot (low curvature), while other directions we have to make small steps■
- In adaptive methods we use a dynamic algorithm to rescale the step size of each parameter (weight)■
- We can think of this as a “regularisation” that makes our basins of attraction more spherical■

# Rescaling Co-ordinates



# AdaDelta

- We want to rescale the coordinates in proportion to the curvature in that direction■
- To estimate the curvature we compute a running average of the squared gradients

$$S_i^g(t+1) = (1 - \gamma)S_i^g(t) + \gamma \left( \frac{\partial L_{\mathcal{B}}(\mathbf{w}(t))}{\partial w_i(t)} \right)^2 \quad \blacksquare$$

- To estimate the relative scale of the weights we compute a running average of the squared weights

$$S_i^w(t+1) = (1 - \gamma)S_i^w(t) + \gamma w_i(t)^2 \quad \blacksquare$$

# AdaDelta

- The AdaDelta algorithm uses the update

$$w_i(t+1) = w_i(t) - \eta \sqrt{\frac{S_i^w(t+1) + \epsilon}{S_i^g(t+1) + \epsilon}} \frac{\partial L_{\mathcal{B}}(\mathbf{w}(t))}{\partial w_i(t)}$$

- Note that we can rescale the weights or rescale the gradients yet the update would be the same—this is one aspect of a covariant algorithm
- Because we are adaptively changing our step size we can use a single step size throughout the optimisation

# Adding in Momentum

- AdaDelta doesn't use momentum (there is an argument this isn't so important as it has “regularised the landscape”) ■
- Nevertheless we can learn a momentum term

$$M_i(t+1) = (1 - \beta)M_i(t) + \beta \frac{\partial L_{\mathcal{B}}(\mathbf{w}(t))}{\partial w_i(t)}$$
$$S_i(t+1) = (1 - \gamma)S_i(t) + \gamma \left( \frac{\partial L_{\mathcal{B}}(\mathbf{w}(t))}{\partial w_i(t)} \right)^2 \quad \blacksquare$$

- These running averages have a lag time which we can remove

$$\hat{M}_i(t+1) = \frac{M_i(t+1)}{1 - (1 - \beta)^t} \quad \hat{S}_i(t+1) = \frac{S_i(t+1)}{1 - (1 - \gamma)^t} \quad \blacksquare$$

# ADAM

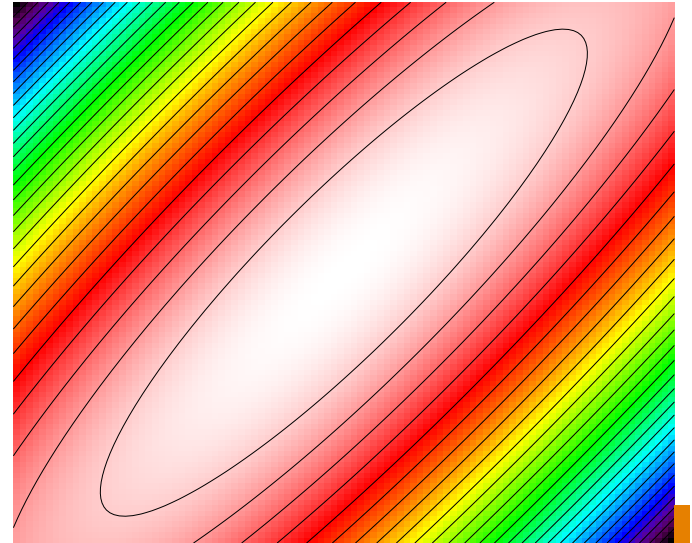
- Adaptive Moment Estimation (Adam) adapts the scale of the parameters, but also uses momentum■
- Update weights

$$w_i(t+1) = w_i(t) - \frac{\eta}{\sqrt{\hat{S}_i(t+1) + \epsilon}} \hat{M}_i(t+1) \blacksquare$$

- ADAM and its variants are very successful in deep learning■
- It is very robust so often results in a network learning which might not learn under standard SGD (with momentum)■

# Covariant Equations

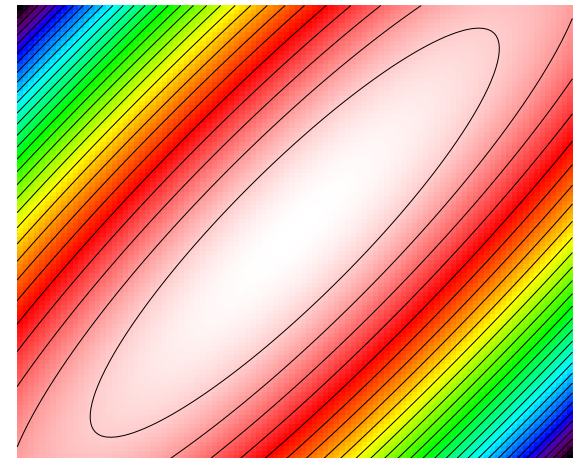
- Vector arithmetic, (matrix multiplication, addition, multiplying by a scale, computing gradients) has a covariant property that it is invariant to the coordinate system■
- That is we can translate and rotate our coordinates and get the same result■
- That is not true of element-wise multiplication of vectors■
- Many ML algorithms do this (including ADAM and adaDelta), but they aren't invariant if we rotate our coordinates■



Can't rescale  
coordinates■

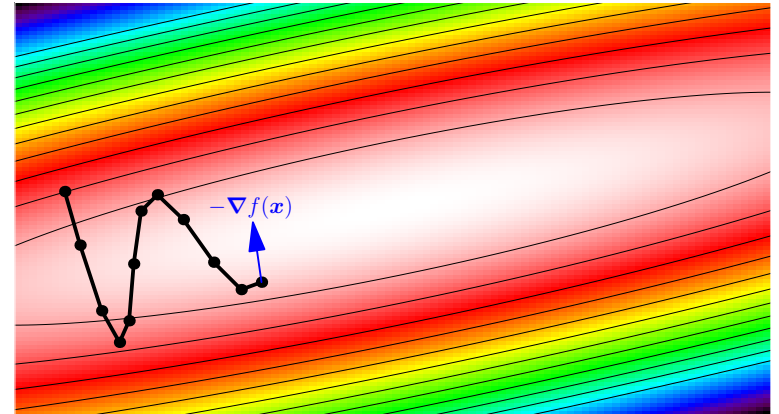
# Correlated Weights

- Correlated weights will lead to skewed valleys■
- Adaptive methods would be more efficient if they were rotationally invariant■
- However, this would slow down the algorithm■
- ADAM is a compromise between speed of implementation and effectiveness■



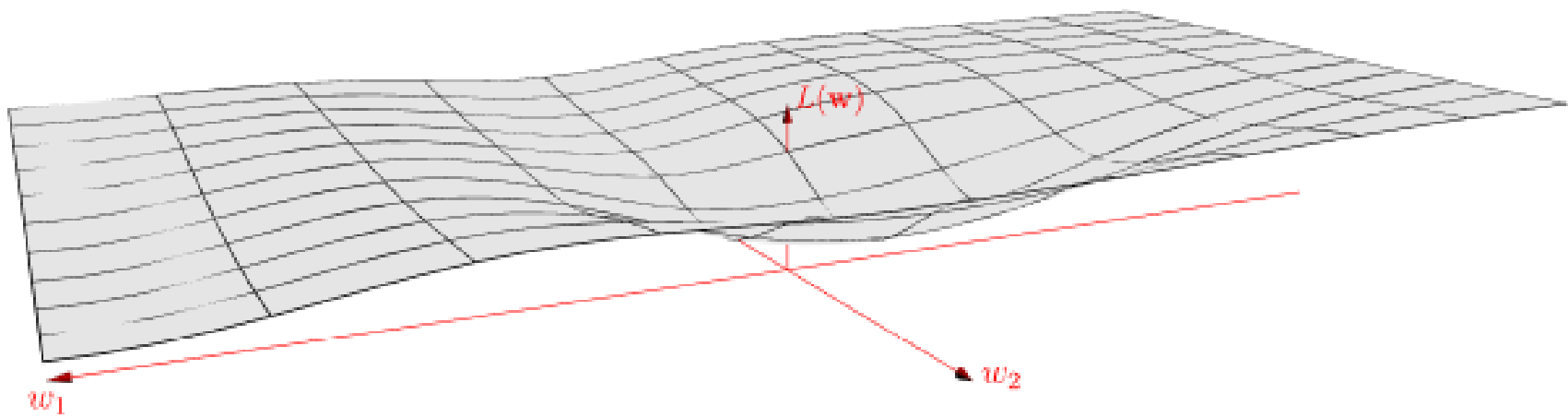
# Outline

1. SGD
2. Momentum
3. **Loss Landscapes**



# Loss Landscape

- A useful concept for understanding optimisation is the loss landscape
- For every value of the weights  $w$  there is some loss associated with it
- Note that this landscape is very high dimensional, so our intuition can be a bit misleading
- The landscape is also huge



# Global and Local Minima

- Our objective is to minimise our loss■
- This would mean finding the global minimum■
- There are no algorithms that are guaranteed to find the global minimum■
- The best we can hope is to find a local minimum by doing gradient descent■
- In practice, our landscapes are so large that we are probably unable to find even a local minimum■

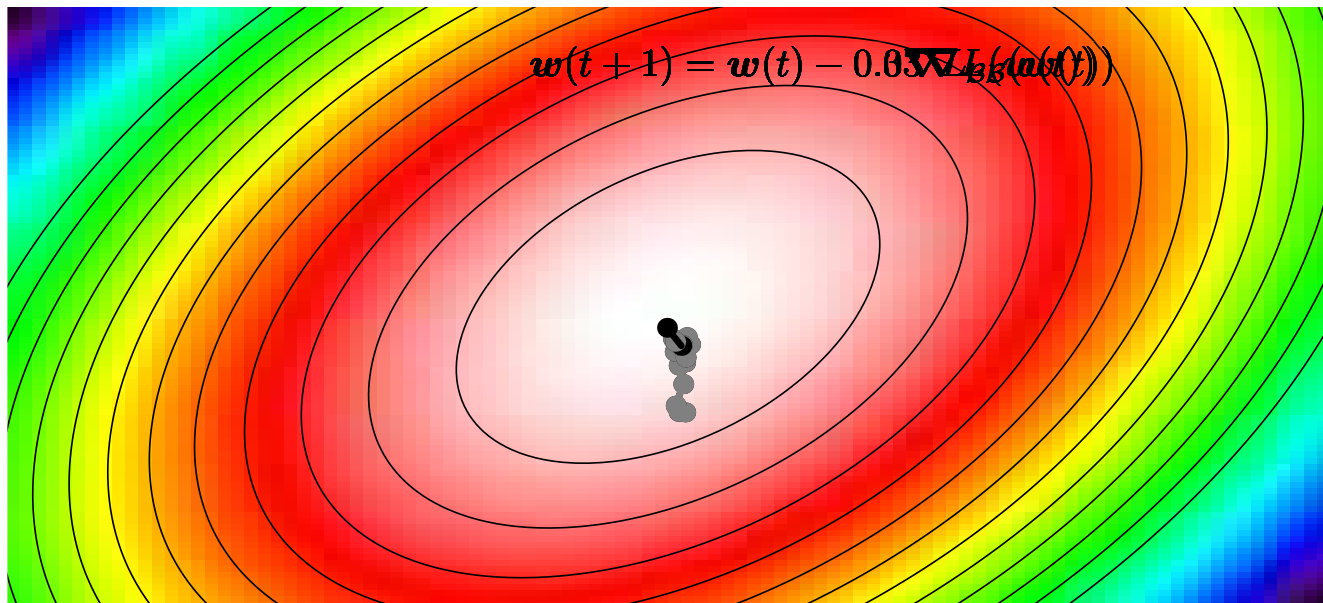
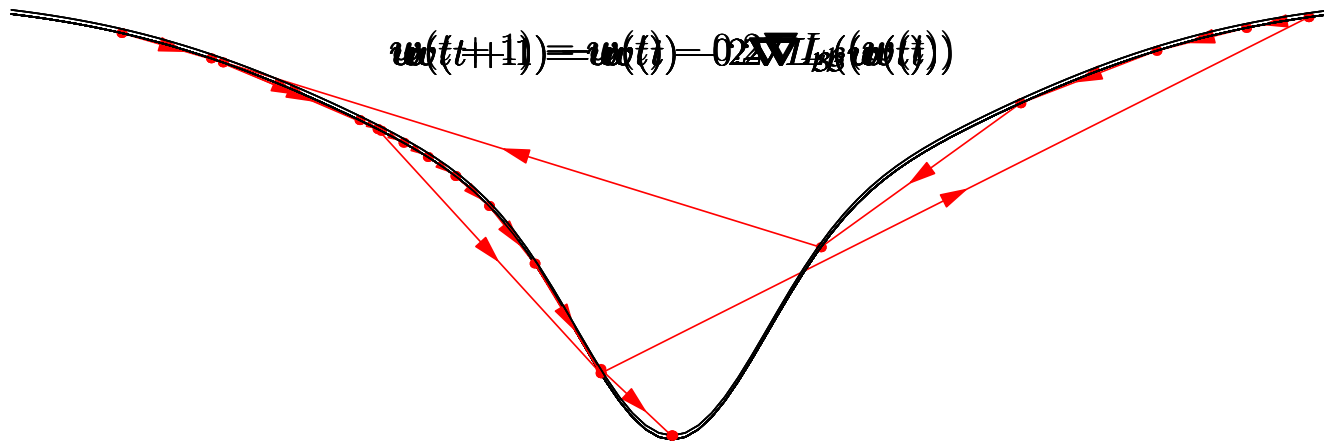
# Noisy Optimisation

- Because we are using mini-batches we aren't actually following the correct gradients■
- If we reduce our step size then our average gradient is closer to the true gradient■
- Thus our optimisation is inherently noisy■
- At least for deep learning there is no evidence that the weights stop changing even after learning for a huge amount of time■

# Valleys in Valleys in Valleys

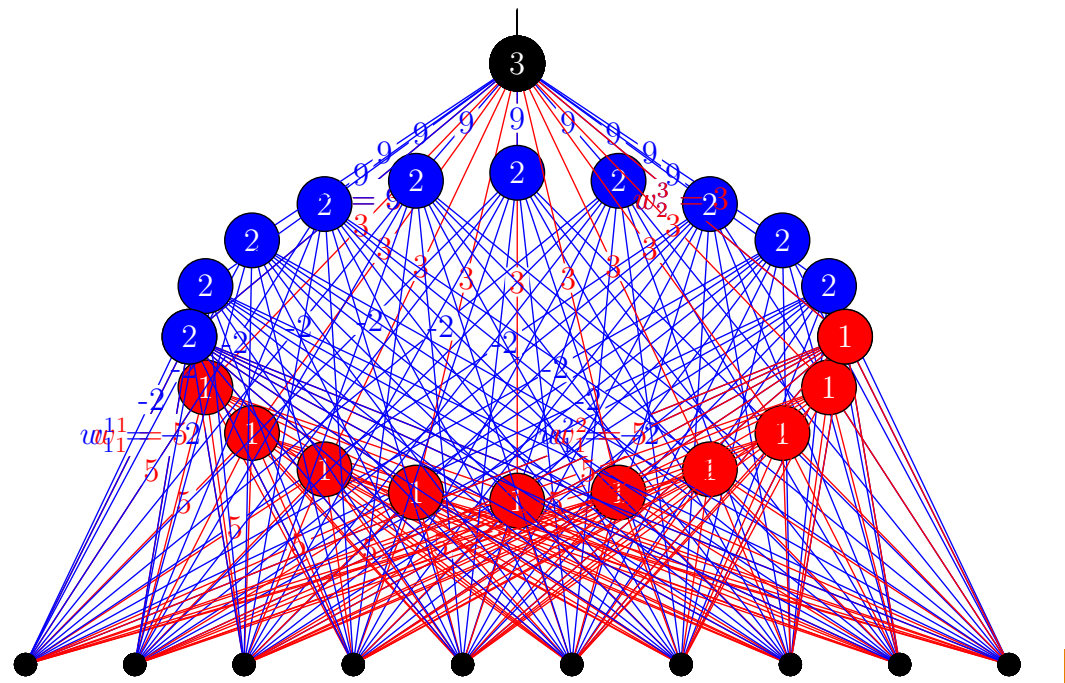
- My view of the loss landscape is that we have valleys inside bigger valleys inside bigger valleys, and so on■
- If our noise is high (we use large step size), we can navigate away from local valleys and move towards the big valley■
- But, we can't explore the small valleys■
- Often when the rate of improvement slows down, practitioners will reduce their learning rate by a factor of 10 and the rate of improvement jumps up■
- Some practitioners cycle through raising and lowering their learning rates which may speed up convergence■

# Reducing the Step Size



# Symmetries

- When training neural networks (deep or shallow) there are typically many symmetries
- Examples come from exchanging the weights of neurons (or filters in CNNs) in two layers



# More Symmetries

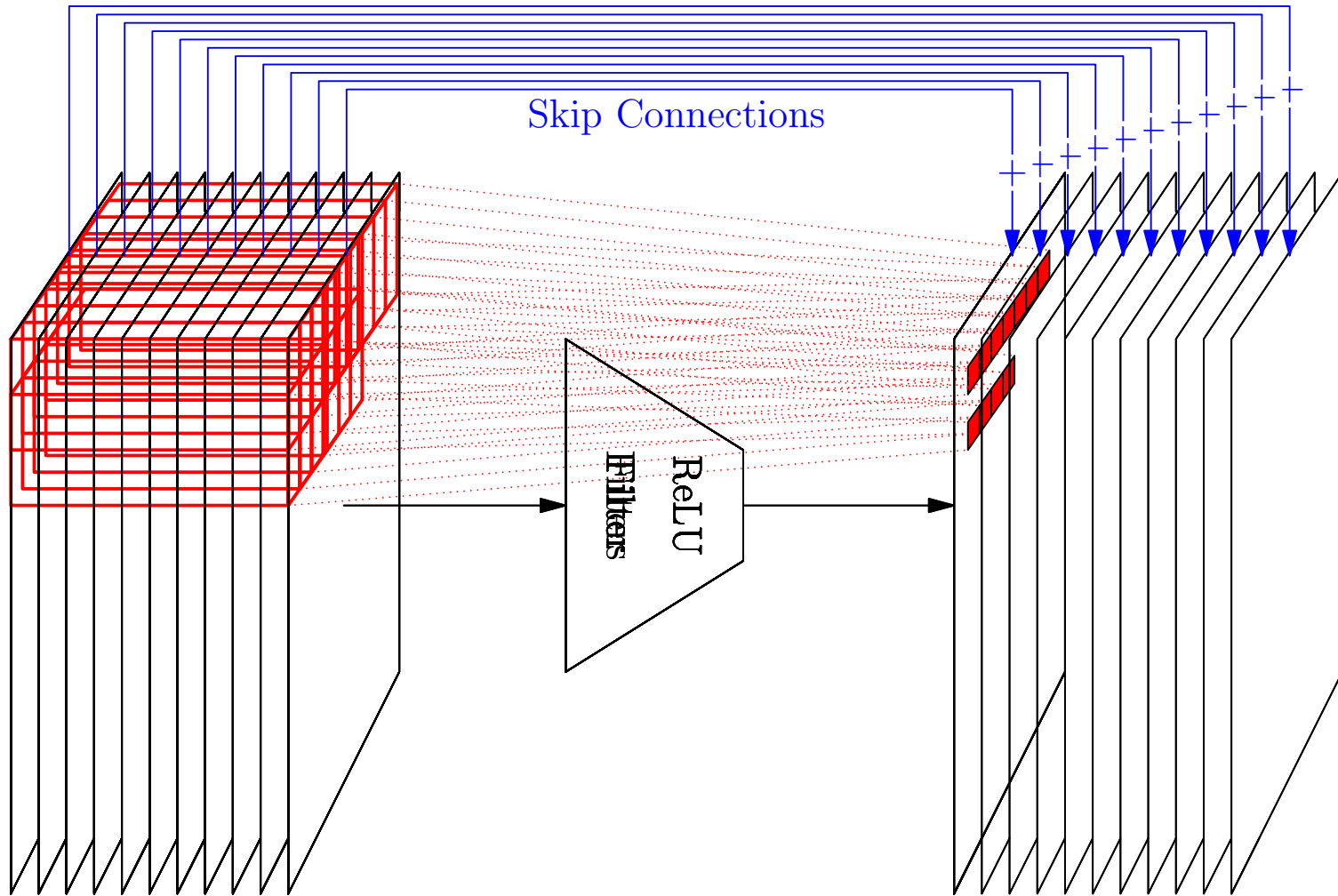
- With  $n$  hidden nodes there are  $n!$  symmetries
- Often if I multiply all the input and output weights to a node by  $-1$  the network will be the same
- There are also continuous symmetries. For linear networks with two layers performing a mapping  $\mathbf{W}_2\mathbf{W}_1$  then for any invertible matrix  $\mathbf{U}$  (where  $\mathbf{U}\mathbf{U}^{-1} = \mathbf{I}$ )

$$\mathbf{W}_2\mathbf{W}_1 = \mathbf{W}_2\mathbf{I}\mathbf{W}_1 = \mathbf{W}_2(\mathbf{U}\mathbf{U}^{-1})\mathbf{W}_1 = (\mathbf{W}_2\mathbf{U})(\mathbf{U}^{-1}\mathbf{W}_1)$$

so a network with weights  $\mathbf{W}'_2 = \mathbf{W}_2\mathbf{U}$  and  $\mathbf{W}'_1 = \mathbf{U}^{-1}\mathbf{W}_1$  will perform the same mapping

- There will be a huge space of equivalent networks

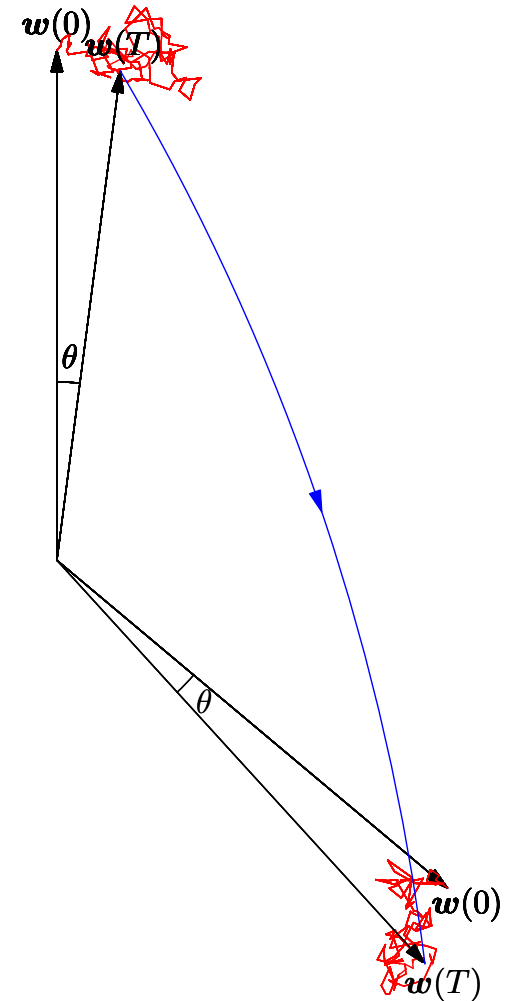
# Skip Connections



- Breaks permutation symmetry. Used in Resnets, transformer, etc.

# Symmetries

- The landscape potentially has a huge manifold with the same losses (neutral network)
- Find solution close to start
- We should think of optimisation more as a process of travelling than a process of arriving
- Of course even if we do minimise our loss we are not guaranteed to minimise our generalisation performance



# Lessons

- SGD together with automatic differentiation has revolutionised machine learning■
- There are a number of techniques to get the step size right■ momentum■ adaptive step size■ ADAM■
- Still need to understand that we are exploring a huge and complex loss landscape■
- What works is problem specific■(as always)■